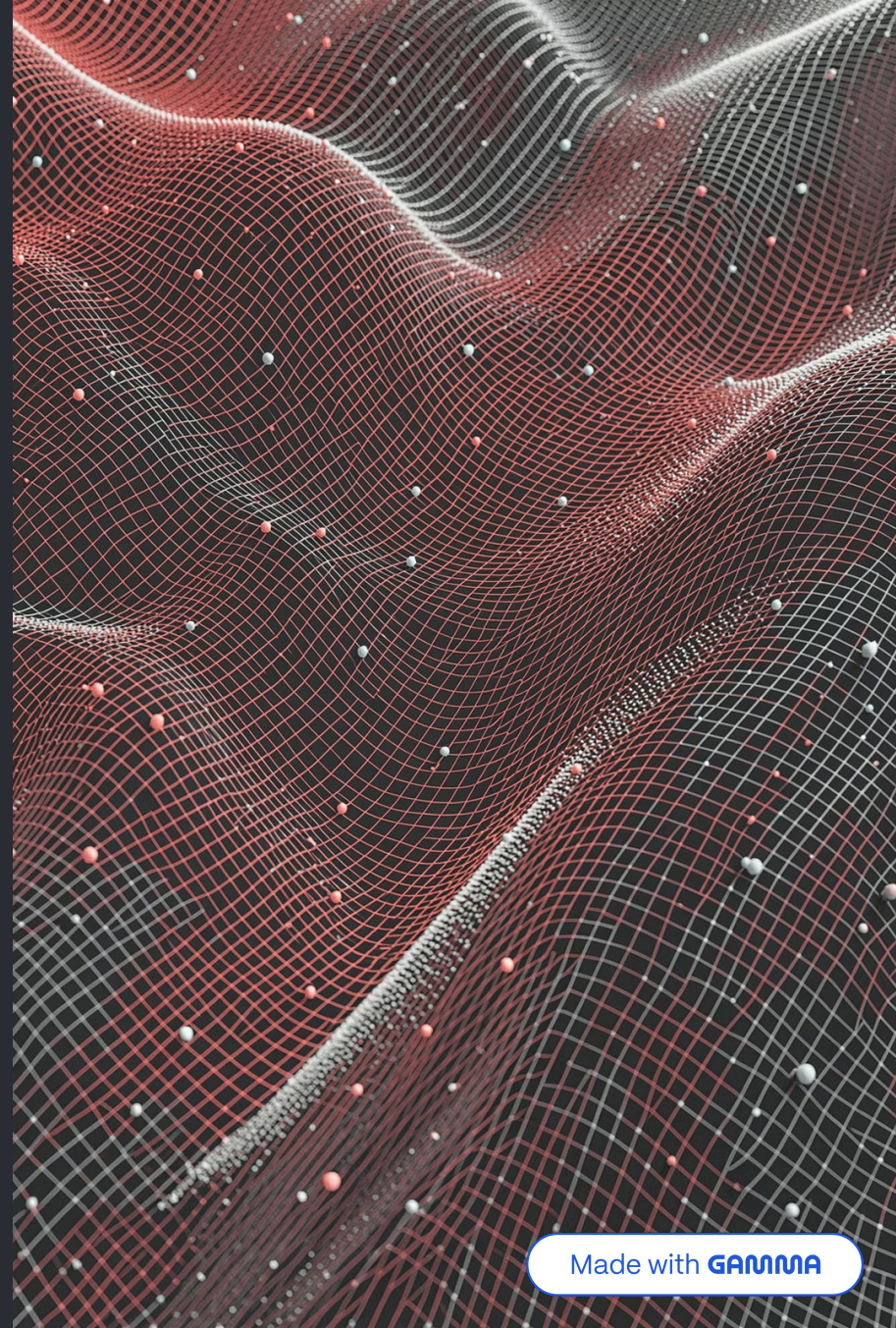
Data Structures in C for Digital Marketing Tools

High-performance solutions for Ad Tech, SEO, and analytics



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Why C in Modern Marketing?

Big Data Reality

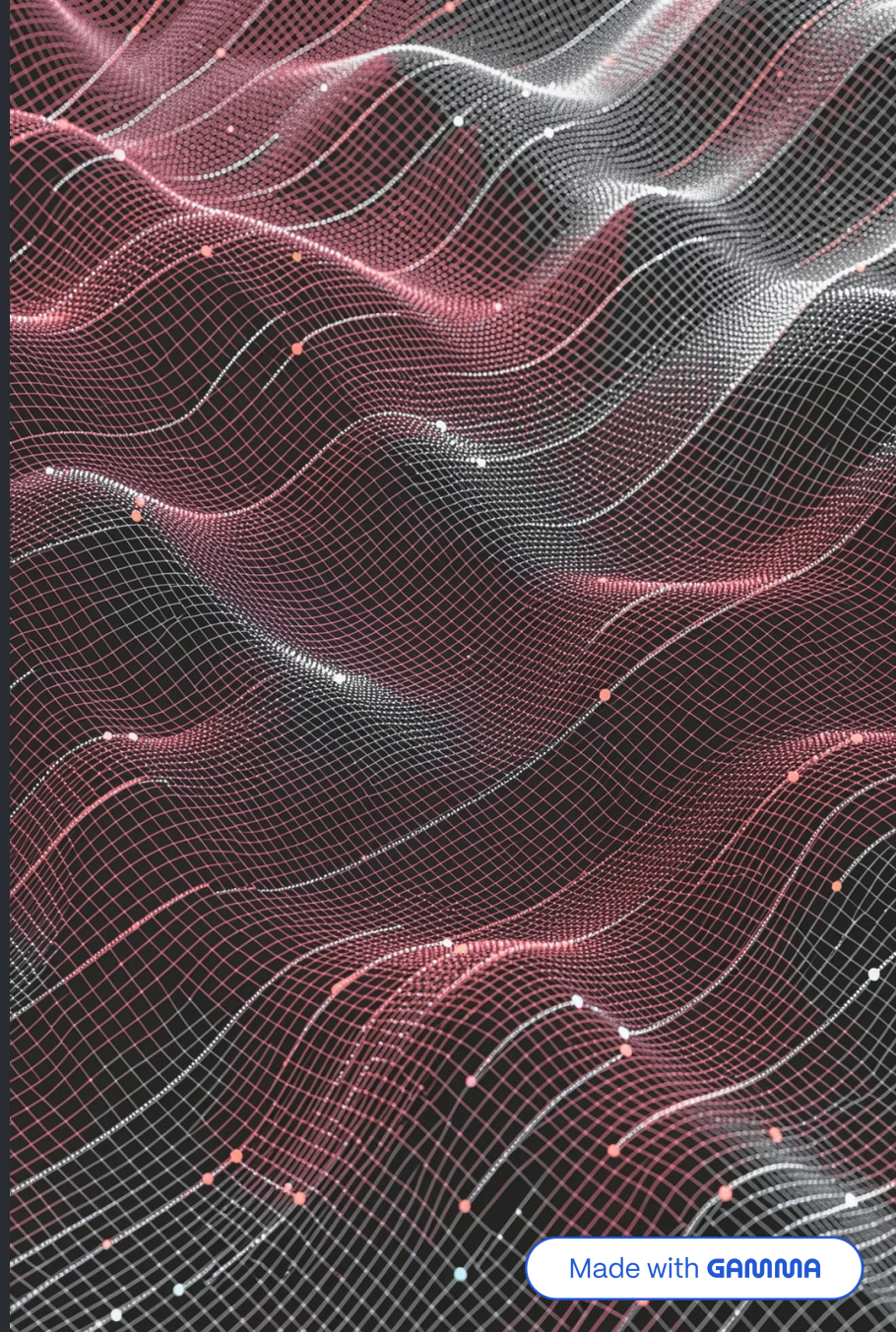
Millions of clicks, impressions, user logs processed daily across platforms

Millisecond Latency

Real-time bidding demands sub-100ms decisions on ad serving

Core Engine Power

C drives ad servers, databases, routers—Python/R handle analysis only



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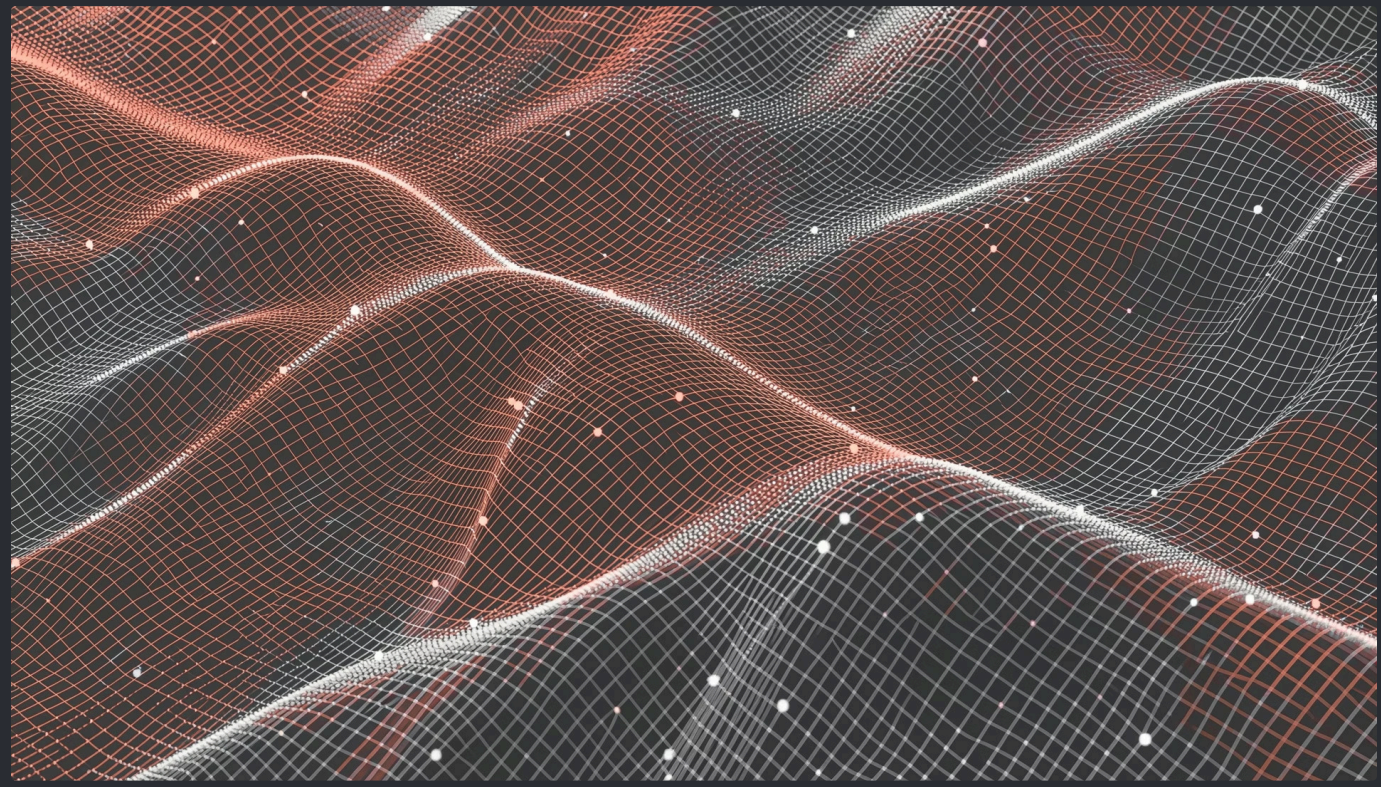
The Core Challenge

Data Volume vs. Processing Speed

- 100,000+ requests per second
- Dynamic keywords, bids, user IDs
- Limited RAM, no memory leaks



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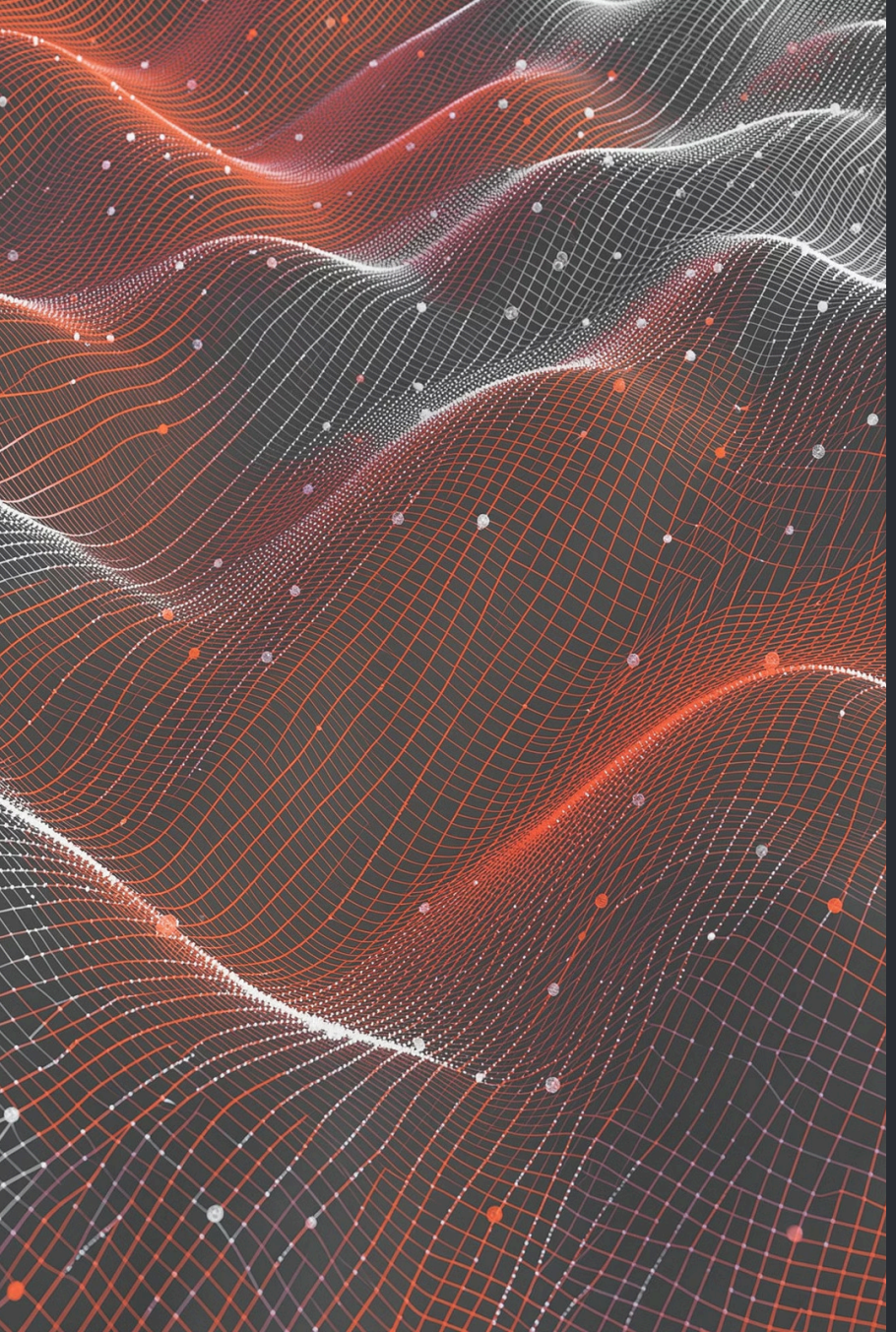

Arrays & Strings: The Foundation

Keyword Management

Storing negative keywords, excluded IPs, ad creatives for rotation

Contiguous Memory

O(1) access time enables fast iteration through millions of entries

Network Buffering

Essential for buffering incoming user data packets at scale

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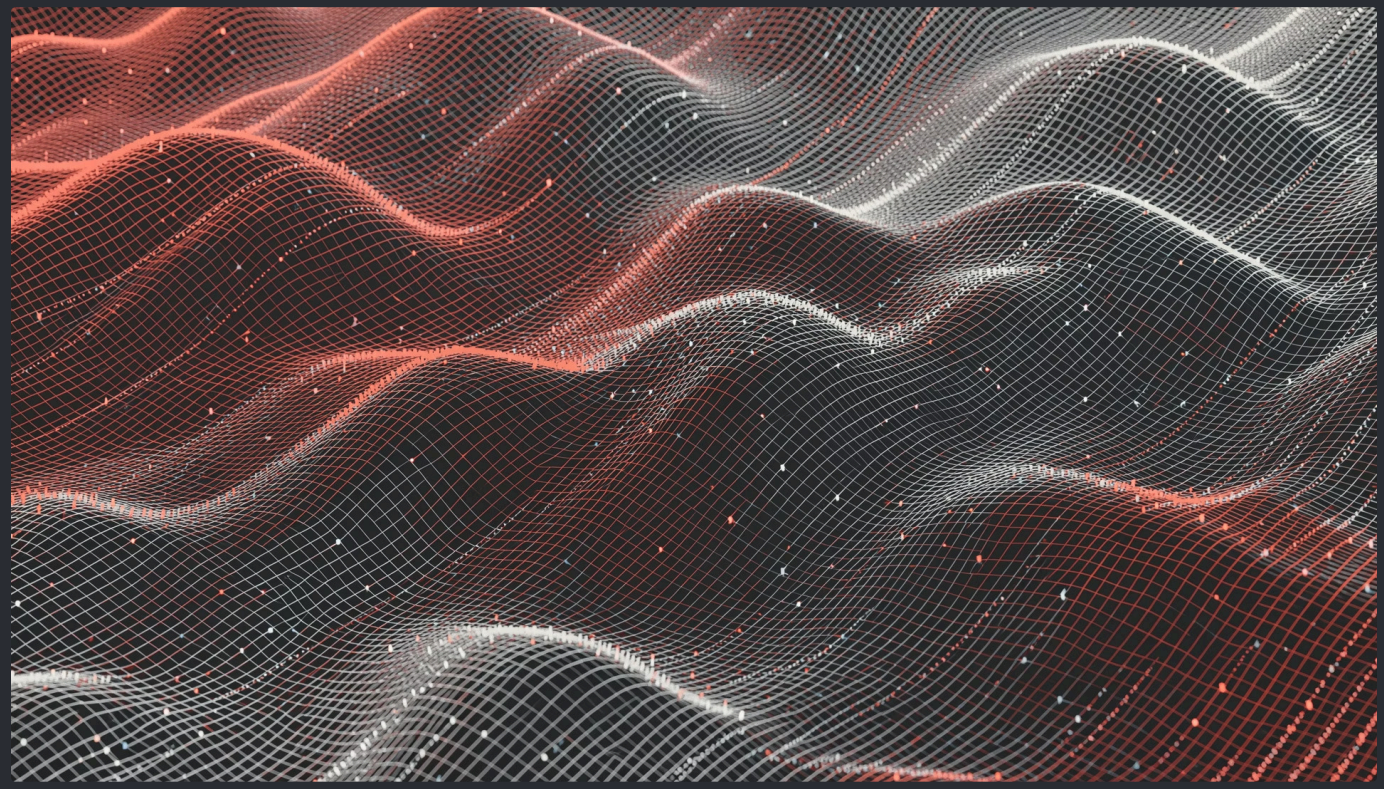
Hash Tables: Real-Time Bidding Key

User Cookies & Session Management

Instant lookup of user IDs across millions of records

Frequency Capping

Track ad view counts: UserID → View Count mapping



Why C? Average O(1) complexity for search, insert, delete—critical for sub-100ms ad server decisions

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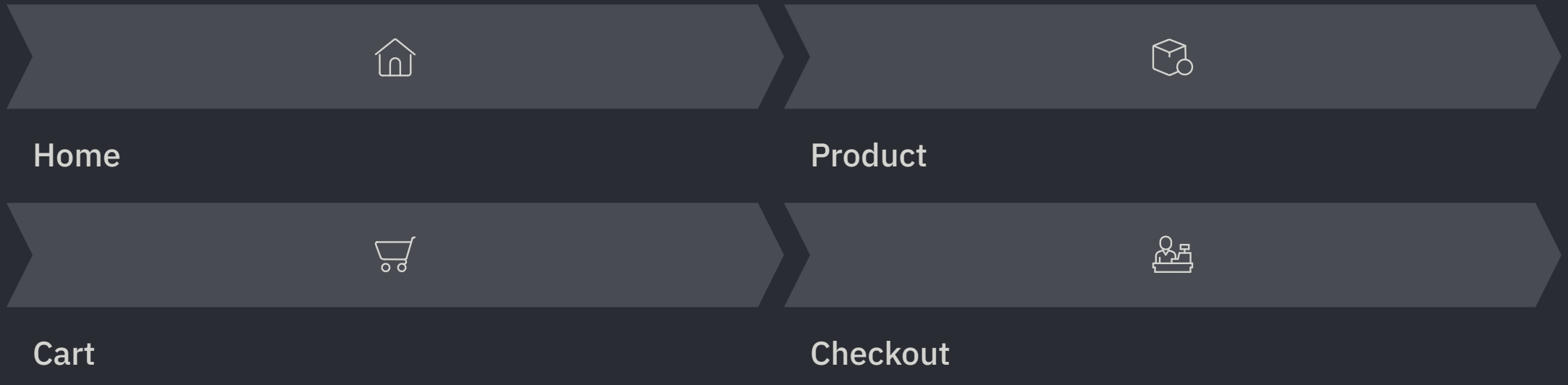
Linked Lists: Dynamic Log Handling



Chronological clickstream data • Dynamic size • Easy insertion/deletion without memory reorganization

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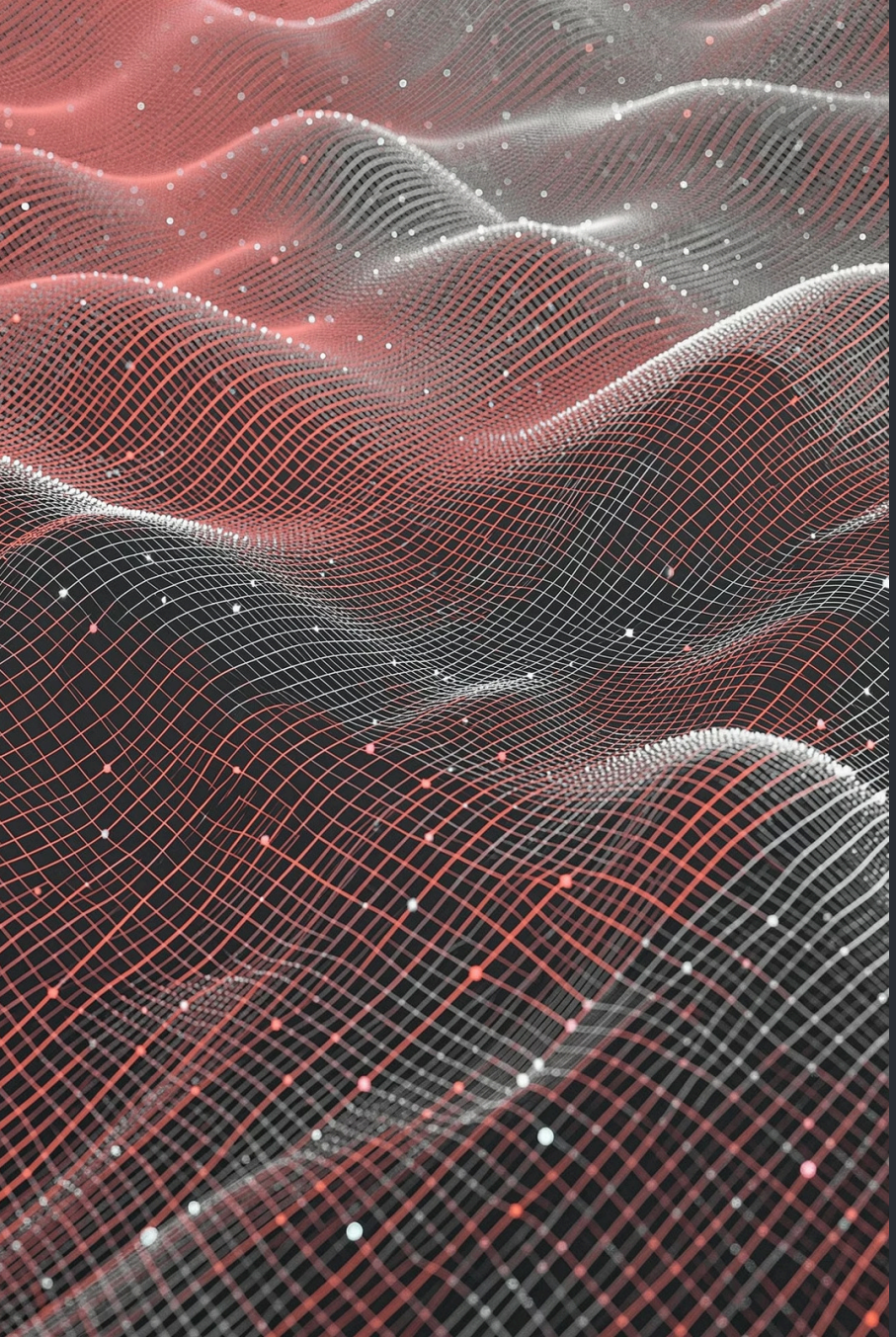

Trees: Auto-Complete & Hierarchies

01

Tries (Prefix Trees)

Power Google Ads auto-complete and search suggestions

02

Binary Search Trees

Store campaign hierarchy: Account → Campaign → Ad Group → Ad

03

Efficient Lookup

Faster than linear arrays for sorted data—O(L) predictive text

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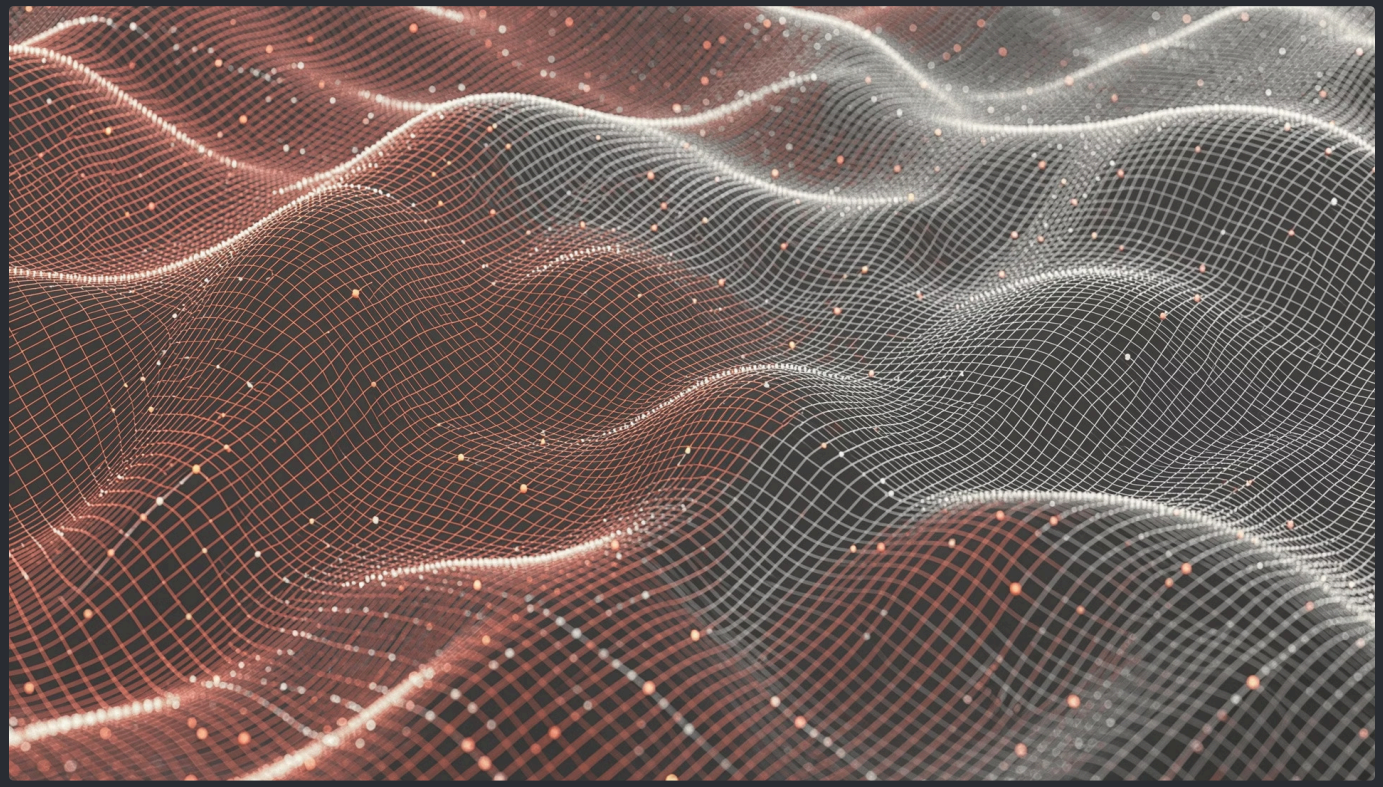
Graphs: Social Network Analysis



Influencer Mapping & Social Connections

- Map user relationships on social platforms
- Identify key influencers via PageRank logic
- Find shortest path between product and buyer

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Code Example: Frequency Capping

```
struct UserView {
  char userID[20];
  int viewCount;
};

int canShowAd(char *id) {
  int index = hashFunction(id);

  if (hashTable[index]->viewCount < 3) {
    hashTable[index]->viewCount++;
    return 1;  // Show Ad
  }
  return 0;  // Block Ad
}
```

This logic runs millions of times per second on production ad servers

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Why C for Marketing Tools?

Performance

No garbage collection pauses—unlike Java/Python

Memory Control

Precise control over every byte for large datasets

Portability

Runs on servers, IoT devices, any hardware

Legacy Libraries

Core marketing analytics built in C/C++

> Digital marketing demands data engineering excellence—master data structures to build scalable, high-performance marketing engines

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